

Risk Management

- ◇ Risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on one or more project objectives such as scope, schedule, cost, and quality.
- ◇ Risk Management is the process of identifying and migrating risk.
- ◇ Risk management is defined as “the systematic application of management practices, policies, and procedures for identifying, analyzing, controlling and monitoring risk.



Why is it important?

- ◆ Risk affects all aspects of your project-your budget, your schedule, your scope, the agreed level of quality, and so on.
- ◆ Increase probability of positive event.
- ◆ Reduce the occurrence of negative events.

How do we manage risk ?

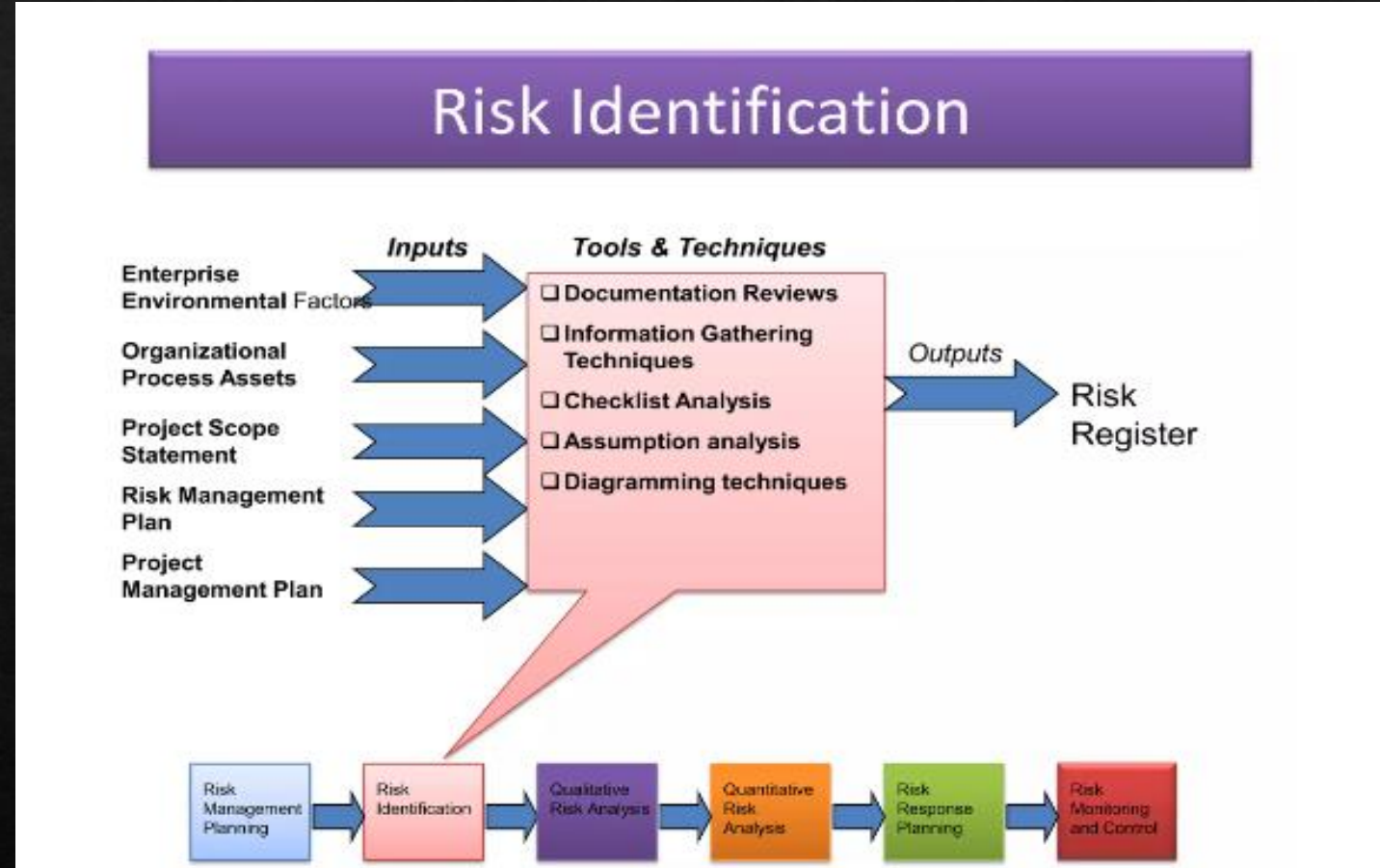
- ◆ To manage risk we use the risk management process
 - Risk Identification
 - Risk analysis
 - Risk response planning
 - Risk monitoring and control

Risk management planning

- ◊ Analysis and decision making to implement risk management according to size and complexity of the project.
- ◊ Understand the general level of risk your project faces
- ◊ Stakeholders will be involved in planning risk management.
- ◊ Risk Management Plan should include:
 - Schedule Risk
 - Cost Risk
 - Quality Risk
 - Scope Risk
 - Resource Risk
 - Customer Satisfaction Risk

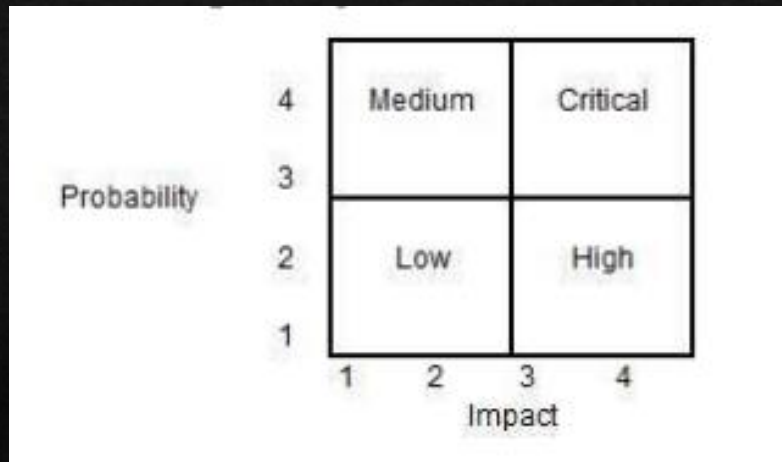
Risk Identification

- ◆ Identify individual risks that might affect your project by making a list
- ◆ Which risk has more probability of affecting the project?
- ◆ Use wide variety of inputs, tools , and technique to identify all the risks to the project.
- ◆ Performed → Throughout the life of the project



Risk Analysis

- ◇ begin putting this list of risks in a **risk register**—a chart that lays out each risk, followed by information like priority level and mitigation plans
- ◇ Helps to identify and manage potential problem
- ◇ Using the matrix, the project manager can categorize the risk into four categories as Low, Medium, High and Critical.
- ◇ The probability of occurrence and the impact on the project are the two parameters used for placing the risk in the matrix categories.



Probability	4	Medium	Critical
	3		
	2	Low	High
	1		
		1 2 3 4	Impact

- ◇ The categorization of risk can be done on the percentage basis , very low(75%) ,here percentage resemble either its probability of occurrence in a project or its impact on the project.

Risk Response Planning

- ◇ assigning each risk a high, medium, or low priority based on the factors you've determined
- ◇ Eliminate threats before they happen.
- ◇ Decrease impact of threat
- ◇ Contingency Plan (Do Something if risk happens)
- ◇ Fallback Plan (DO something is contingency plans are not effective)

Negative Risk or Threats	Positive risk or Threats
Avoid Transfer Mitigate (minimize) Accept	Exploit Share enhance

Monitor and Control Risk

- ◇ set up a process to monitor each risk as your project begins
- ◇ Risk Assessment → Schedule regularly to identify new risk
- ◇ Risk Audit → Examine the effectiveness of planned risk response
- ◇ Trend Analysis → Monitor overall project performance

What are the benefits?

- ◆ Effective use of resources
- ◆ Promoting continuous improvement
- ◆ Fewer Shocks and failures
- ◆ Strategic business planning
- ◆ Raised awareness of significant risks.
- ◆ Quick grasp of new opportunity
- ◆ Enhancing Communication
- ◆ Reassurance stakeholders
- ◆ Recognition of responsibility

Categories of Risk

- ◇ Project Risk
- ◇ Technical Risk
- ◇ Business Risk

Project Risk

- ◆ A project risk is any unforeseen thing that might — or might not — occur during a project. A risk isn't necessarily negative; it's just an event where the outcome is uncertain. As such, a project risk can have either a negative or positive effect on the project's objectives.
- ◆ Most common Project Risk
 - **Cost Risk:** It is the risk that the project will cost more than the budget allocated for it. the most common project risk, cost risk is due to poor budget planning, inaccurate cost estimating
 - **Schedule Risk:** Schedule risk is the risk that activities will take longer than expected, and is typically the result of poor planning. It's closely related to cost risk, because slippages in schedule typically increase costs and also delay the outcome of the project.
 - **Performance Risk:** Performance risk is the risk that the project will fail to produce results consistent with project specifications. A project team can deliver the project within budget and schedule and still fail to produce the results and benefits

Business Risk

- ◆ Business risks are uncertain factors, internal or external, that threaten the financial health of an organization.
- ◆ Examples of external business risks would be natural disasters or cyber attacks. Internal business risks are threats that come from within the company, such as falling out of compliance, having too much debt, or labor disputes.
- ◆ Top business risk are :
 - **Market risk:** Building an excellent product or system that no one really wants.
 - **Sales risk:** Building a product that the sales force does not understand how to sell.
 - **Management Risk:** losing the support of senior management due to change in focus or change

Technical Risk

- ◆ Threaten the quality and timeliness of the software to be produced.
- ◆ If the technical risk become a reality , implementation may become difficult or impossible
- ◆ Some reasons for Technical risks –
 - ◆ Frequent changes in requirement
 - ◆ Less use of future technologies
 - ◆ Less number of skilled employee
 - ◆ High complexity in implementation
 - ◆ Improper integration of modules

Framework for dealing with risk(Risk Management Framework)

- ◇ The Risk Management Framework is a template and guideline used by **companies to identify, eliminate and minimize risks.**
- ◇ It was originally developed by the National Institute of Standards and Technology to help protect the information systems of the United States government.
- ◇ An effective risk management framework seeks to **protect an organization's capital base and earnings without hindering growth.**
- ◇ Adopting a risk management framework that incorporates best practices into the firm's risk culture can be the foundation of a company's financial future.

5 Component of RMF

- ◆ When developing a risk management framework, at least five critical components must be considered.
 - Risk identification,
 - Risk measurement and assessment,
 - Risk mitigation (Minimization),
 - Risk reporting and monitoring,
 - Risk governance

Risk Identification

- ◆ The first component in implementing the Risk Management Framework is to identify the risks that the organization faces. These might include strategic, legal, operational and privacy risks.
- ◆ It is important to note that risk identification is not a one-time process. The risks that an organization faces tend to change over time, so risk assessments will need to be performed on a periodic basis.
- ◆ After listing all possible risks, the company can then select the risks to which it is exposed and categorize them into **core and non-core risks**.
 - ◆ Core risks are those that the company must take in order to drive performance and long-term growth.
 - ◆ Non-core risks are often not essential and can be minimized or eliminated completely.

Risk measurement and assessment,

- ◆ For each risk identified by the process mentioned above, organizations will need to create a detailed risk profile, and assign a score to each risk based on their potential impact. Risk assessments should be carried out at regular intervals.

In some cases, risk measurement might be based on something as simple as how much capital could potentially be lost as a result of the risk. However, in other cases, measuring the potential impact of a risk might be far more difficult.

Risk mitigation,

- ◆ Once a thorough risk assessment has been carried out, organizations will need to establish a plan for mitigating these risks, which will be prioritized based on their risk score.
- ◆ Part of this process involves coming up with mitigation strategies, such as cyber insurance. For example, if an organization identifies cybersecurity risks that need to be dealt with, then it may choose to integrate security controls into its development lifecycle.

Risk reporting and monitoring,

- ◆ Organizations must periodically review their risk identification, assessment, and mitigation strategies to ensure that they are effective, and produce reports that highlight any areas that need improvement.
- ◆ This essentially means regularly reexamining the risks in order to make sure that the risk mitigation strategies the organization has adopted are having the desired effect.

Risk governance

- ◆ Implement all of the risk management steps defined above.
- ◆ Risk governance involves defining the roles of all employees, segregating duties, and assigning authority to individuals, committees, and the board for approval of core risks, risk limits, exceptions to limits, and risk reports, and also for general oversight.

Risk Management Framework Steps

Step	Description
Prepare	➤ focuses on getting the organization ready to adopt a formalized risk management strategy
Categorize	➤ organizations begin assessing the risks that have been identified. ➤ This may mean assessing the impact of the various risks and prioritizing the risks that need to be addressed.
Select	➤ involves <u>choosing the controls</u> that will be used to protect affected systems to minimize or mitigate the risks that have been identified.
Implement	➤ This is where the selected controls are put into place in an effort to head off risks that might exist.
Assess	➤ It seeks to determine whether the selected controls were implemented correctly and if those controls are delivering the desired result ➤ This means making sure any mechanisms that have been implemented are reducing risks in a quantifiable way without accidentally introducing new risks in the process.
Authorize	➤ the authorize phase is more of an overview by senior members of the organization who are looking to make sure that risk mitigation strategies are working

Evaluating risk to the schedule

- ◇ not all risks can be eliminated - even those that are classified as avoidable or
- ◇ By identifying and categorizing those risks, and in particular, their likely effects on the duration of planned activities, we can assess what impact they are likely to have on our activity plan.
- ◇ Two methods for assessing the effects of these uncertainties on the project schedule
 - ◇ Risk Matrix

Severity

- ◆ **Severity** is the amount of damage or harm a hazard could create and it is often ranked on a four point scale as follows:
 - ◆ **Catastrophic - 4:** Operating conditions are such that human error, environment, design deficiencies, element, subsystem or component failure, or procedural deficiencies may commonly cause death or major system loss, thereby requiring immediate cessation of the unsafe activity or operation.
 - ◆ **Critical - 3:** Operating conditions are such that human error, environment, design deficiencies, element, subsystem or component failure, or procedural deficiencies may commonly cause severe injury or illness or major system damage thereby requiring immediate corrective action.
 - ◆ **Marginal - 2:** Operating conditions may commonly cause minor injury or illness or minor systems damage such that human error, environment, design deficiencies, subsystem or component failure, or procedural deficiencies can be counteracted or controlled without severe injury, illness, or major system damage.
 - ◆ **Negligible - 1** Operating conditions are such that personnel error, environment, design deficiencies, subsystem or component failure, or procedural deficiencies will result in no, or less than minor, illness, injury, or system damage.

Probability

- ◇ **Probability** is the likelihood of the hazard occurring and it is often ranked on a five point scale:
 - ◇ **Frequent - 5:** Likely to occur often in the life of an item
 - ◇ **Probable - 4:** Will occur several times in the life of an item
 - ◇ **Occasional - 3:** Likely to occur sometime in the life of an item.
 - ◇ **Remote - 2:** Unlikely but possible to occur in the life of an item.
 - ◇ **Improbable - 1:** So unlikely, it can be assumed an occurrence may not be experienced.

How to Use the Risk Assessment?

- ◇ The Risk Assessment values are determined by multiplying the scores for the Probability and Severity values together.
- ◇ The higher the risk assessment, the greater the overall risk for the project.
- ◇ This method helps balance the weight of severity and probability

Cont....

		Severity			
		Catastrophic: 4	Critical: 3	Moderate: 2	Marginal: 1
Probability	Frequent: 5	High - 20	High - 15	High - 10	Medium - 5
	Probable: 4	High - 16	High - 12	Serious - 8	Medium - 4
	Occasional: 3	High - 12	Serious - 9	Medium - 6	Low - 3
	Remote: 2	Serious - 8	Medium - 6	Medium - 4	Low - 2
	Improbable: 1	Medium - 4	Low - 3	Low - 2	Low - 1